



UNIT TEST RAJENDRA VIDYALAYA

4 Pages

II-VIII

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Standard : 8'A Date : 02/05/26 Roll No. : 46

Q1)

Ans a) Proton

Ans b) Atomic number

Ans c) Neutron

Ans d) Nucleus

Ans e) anion

Q2)

Ans → Two main features of Rutherford's atomic model are:-

- Most of rays passed straight through the foil without any deflection
- A few of them bounced back.

Q3)

Ans → Atomic number = 4

Mass number = 4 Number of protons + Number of neutrons
= 4 + 5

$8 - 4 = 4$

The atom is Beryllium and the symbol is ${}^9_4\text{Be}$.

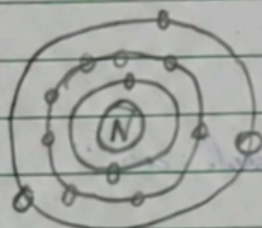
Q4)

Ans → The number of ^{max} protons and the number of neutrons are same but the atomic number is different it is known as isotopes.
The three isotopes of hydrogen are H^1 , H^2 , H^3

Q5)

Ans → $^{13}_{13}Al^{27}$

Atomic diagram



Mass number = 27

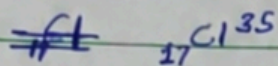
Atomic number = 13

Number of protons =

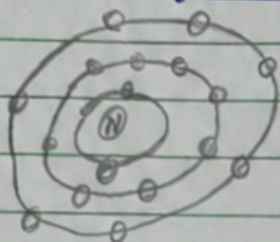
$$\begin{aligned}\text{Number of neutrons} &= A - Z = \text{Mass number} - \text{Atomic number} \\ &= 27 - 13 \\ &= 14\end{aligned}$$

Number of protons = 13

Electronic configuration of Aluminium is = 2, 8, 3



Atomic diagram



Mass number = 35

Atomic number = 17

$$\begin{aligned}\text{Number of neutrons} &= A - Z = \text{Mass number} - \text{Atomic number} \\ &= 35 - 17 \\ &= 18\end{aligned}$$

Number of protons = 17

Electronic configuration of Chlorine is = 2, 8, 7